

1. Administrative & Study Identification

Full Study Title: A Phase IIIb, Global, Multicenter, Randomized, Visual Assessor-Masked Study of the Efficacy, Safety, and Pharmacokinetics of a 36-Week Refill Regimen for the Port Delivery System With Ranibizumab in Patients With Neovascular Age-Related Macular Degeneration (Velodrome) **Study Title:** A Study of the Efficacy, Safety, and Pharmacokinetics of a 36-Week Refill Regimen for the Port Delivery System With Ranibizumab in Patients With Neovascular Age-Related Macular Degeneration (Velodrome)
Short Title/Acronym: Velodrome **Protocol Number:** WR42221 **NCT Number:** NCT04657289 **Posting ID:** 125317155104120619 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting)
Sponsor Name / Company: Hoffmann-La Roche (F Hoffmann-La Roche Ltd) **Investigational Product:** Port Delivery System (PDS) with ranibizumab (100 mg/mL) **Drug Name:** ranibizumab **ATC Code:** S01LA04
Trade Names: Lucentis, Susvimo, Cimerli, Byooviz **Phase of Development:** Phase 3 (PHASE3) / Phase IIIb **Medical Monitor:** Head, EU/ROW Regulatory Affairs Pharmaceutical Division, PDR / Roche Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy, safety, and pharmacokinetics of the Port Delivery System with ranibizumab (PDS) 100 mg/mL delivered every 36 weeks (Q36W) compared with every 24 weeks (Q24W) in patients with neovascular age-related macular degeneration (nAMD). **Secondary Objectives:** To evaluate overall change in vision (Best-Corrected Visual Acuity, BCVA) over time, change from baseline in center point thickness (central foveal thickness), the safety profile (adverse events), and the proportion of participants who do not require supplemental ranibizumab intravitreal injections.

2.2 Background & Rationale To address the heavy treatment burden of frequent anti-VEGF injections in patients with neovascular age-related macular degeneration (nAMD). While standard intravitreal injections (administered every 1-2 months) are effective, the requirement for frequent clinic visits creates a substantial burden on patients, caregivers, and healthcare systems. The Port Delivery System (PDS) is an innovative intraocular implant designed to continuously deliver ranibizumab, potentially extending the interval between treatments to 24 or 36 weeks while maintaining visual and anatomical outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Q24W refill regimen) **Blinding:** Visual Assessor-Masked (Visual acuity examiners are masked to treatment assignment) **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **\$N\$:** 451 (Targeted Enrollment) **Number of Sites:** Global, Multicenter **Estimated Study Start:** 14-Jul-2021 **Estimated Primary Completion:** 07-Jan-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 50 years at the time of signing Informed Consent Form. 2. Initial diagnosis of neovascular age-related macular degeneration (nAMD) within the last 9 months prior to the screening visit. 3. Previously treated with at least three anti-VEGF intravitreal injections for nAMD per standard of care within 6 months prior to the screening visit, and demonstrated a clinical response to anti-VEGF treatment. 4. BCVA of 34 letters (approximate 20/200 Snellen equivalent) or better.

4.2 Exclusion Criteria 1. Any concurrent intraocular condition that would require surgical intervention during the study to prevent or treat visual loss. 2. Active intraocular inflammation, infectious conjunctivitis, keratitis, scleritis, or endophthalmitis in either eye. 3. History of vitreous hemorrhage or rhegmatogenous retinal detachment in the study eye. 4. History of stroke or myocardial infarction within the last 3 months prior to informed consent. 5. Prior treatment with brodalumab at any time prior to the screening visit in either eye.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ranibizumab 100 mg/mL. Participants receive a surgically implanted PDS on Day 1. Eligible patients are subsequently randomized to receive PDS refill-exchange procedures on a fixed interval of either every 36 weeks (Q36W) or every 24 weeks (Q24W). **Route of Administration:** Intraocular (surgically implanted refillable port delivery system). **Duration of Treatment:** Up to 72 weeks of primary follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard optical correction and routine general medical treatments not contraindicated for ocular health. **Prohibited:** Supplemental anti-VEGF treatments outside the defined study rescue protocol, concurrent intraocular surgeries, and systemic treatments impacting AMD progression.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Baseline BCVA, OCT | | **Baseline** | Day 1 | Surgical implantation of PDS (pre-filled with ranibizumab) | | **Randomization** | Week 24 | Randomization to Q24W or Q36W refill intervals (if disease activity criteria are met) | | **Interim** | Weeks 24 - 68 | Scheduled Refills (per randomization), Safety & Efficacy OCT/BCVA Assessments | | **End of Study** | Week 72 | Final BCVA assessment, Final OCT, Safety Follow-Up, Optional transition to Portal extension study |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in Best-Corrected Visual Acuity (BCVA) from baseline, averaged across weeks 68 and 72.

Measurement Method: Visual acuity assessment using the Early Treatment Diabetic Retinopathy Study (ETDRS) chart starting at a distance of 4 meters.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in BCVA score over time. 2. Percentage of participants with BCVA score of 69 letters (approximate 20/40 Snellen equivalent) or better averaged over Weeks 68 and 72. 3. Percentage of participants with BCVA score of 69 letters (approximate 20/40 Snellen equivalent) or better over time. 4. Proportion of participants who do not require supplemental ranibizumab intravitreal injections.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of systemic and ocular AEs throughout the study duration (e.g., vitreous hemorrhage, retinal detachment, implant dislocation). **Serious**

Adverse Events (SAEs): Reported within 24 hours to the Sponsor/Regulatory bodies. **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms managed by Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for all treatment-emergent adverse event evaluations. **Sample Size Justification:** Target enrollment of 451 patients, powered to evaluate non-inferiority/equivalence of the Q36W regimen to the Q24W regimen regarding visual acuity outcomes. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) and specific masking protocols to handle missing ETDRS letter scores.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine remote and onsite monitoring visits focusing on the surgical procedural adherence, accurate BCVA masking, and refill-exchange procedures. **Audit Trail:** eCRF locking, source document verification, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: WR42221 **Registry Number:** NCT04657289 **Posting ID:** 125317155104120619 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** Velodrome **Official Regulatory Title:** A Phase IIIb, Global, Multicenter, Randomized, Visual Assessor-Masked Study of the Efficacy, Safety, and Pharmacokinetics of a 36-Week Refill Regimen for the Port Delivery System With Ranibizumab in Patients With Neovascular Age-Related Macular Degeneration (Velodrome)

12. Clinical Framework (nAMD Specific)

Primary Condition (Preferred UMLS Name): Neovascular Age-Related Macular Degeneration (nAMD) **Disease Areas:** Neovascular Age-related Macular Degeneration (nAMD) **MeSH Code:** D008268 **HPO Code:** HP:0000608 **SNOMED ID:** 422338006 **Diagnosis Threshold:** Confirmed presence of wet AMD requiring anti-VEGF intervention **Medical Methodology:** Intraocular Anti-VEGF sustained release **Optimization Target:** Sustained visual acuity via PDS refill cycles (Q24W or Q36W)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for surgical PDS implantation and scheduled refill-exchange procedures. **Education Protocol (HCP):** Extensive training on surgical implantation technique and the customized refill-exchange procedure. **Education Protocol (Patient):** Post-operative care, recognition of signs of intraocular infection/inflammation, and adherence to visit schedules. **Quality Audit Mechanism:** Strict masking protocols restricting visual acuity assessors from accessing participant study eye assignments or visit types.

14. Observation & Follow-Up Schedule

Total Duration: Up to 72 primary follow-up weeks **Onsite Visit**
Cadence: Implantation at Day 1, with major assessment and refill milestones at Weeks 24, 36, 48, 60, and 72 **Remote Monitoring**
Frequency: Supplemental virtual checks per site protocols
Checklist Review Interval: Continuous tracking of rescue treatment necessity and ocular health at each visit

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					